
Reasoning Depth versus Orchestration in Mathematical AI Harnesses

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Abstract

1 Mathematical AI systems increasingly wrap language models in harnesses that
2 allocate work among model calls, critics, tools, and persistent state. We ask whether
3 a fixed inference budget is better spent on longer coherent reasoning or external
4 orchestration. We compare direct long reasoning, a multi-agent harness, and Elenx,
5 a reasoning-heavy system that preserves durable state and independently verifies
6 candidate proofs. Our evaluation measures verified solutions as a function of
7 total dollars and decomposes spend into reasoning, coordination, verification, and
8 retries.

9 1 Introduction

10 Accuracy-only comparisons hide where inference budget goes. Elenx keeps search inside long
11 reasoning episodes, while a small kernel records exact candidates, calls, and verifier-bound verdicts.
12 We ask whether this reasoning depth is a better use of budget than adding orchestration. Our
13 contributions are a matched-cost protocol, an auditable comparison, and a budget decomposition.

14 2 Systems and hypothesis

15 We compare a direct long reasoning call, the author’s earlier multi-agent harness, and Elenx. All
16 arms receive the same base model, statement, tools, verifier, and dollar budget. We hypothesize that
17 coherent reasoning can improve the capability–cost frontier on difficult research-style problems, but
18 may be dominated when the base model is already strong.

19 3 Evaluation

20 Tasks are split into development and held-out sets. A success is a complete proof or disproof accepted
21 by an independent verifier. We run paired trials at fixed dollar budgets and report verified tasks solved,
22 cost, reasoning tokens, coordination tokens, calls, retries, and elapsed time. Every paid leaf records
23 provider, model, reasoning level, usage, and billed cost.

24 4 Initial evidence

25 An Elenx campaign on IMO 2011 Problem 2 contains 26 model calls, 13 tool calls, four candidate
26 proofs, and four independent verdicts. Three candidates were rejected with localized objections to
27 load-bearing lemmas. The fourth received PASS after a self-contained proof covering all cardinalities.
28 This is an audited case study, not yet a comparative estimate.

29 5 Results and ablations

Table 1: Matched-budget results (to be completed).

System	Budget	Verified	Cost/pass
Direct reasoning	B_1	–	–
Multi-agent harness	B_1	–	–
Elenx	B_1	–	–

30 We ablate continuation, durable state, independent verification, and long single-episode reasoning. A
31 system is better when it solves at least as many tasks for no greater cost.

32 6 Discussion

33 A negative result would identify orchestration overhead as a measurable design cost; a positive result
34 would support long coherent reasoning as a useful harness primitive. Claims will remain scoped to
35 the tested benchmark, model snapshot, and verifier.

36 References

37 [1] Anonymous. Elenx kernel specification, 2026.